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Research Article

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Rational Design and Computational Validation of a Multi-Epitope Vaccine Candidate Targeting Alto Paraguay Hantavirus Nucleocapsid Protein

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ABSTRACT

Hantaviruses represent a significant public health threat with case fatality rates reaching 35-50% in severe infections. No licensed vaccine currently exists for hantavirus prevention. Alto Paraguay hantavirus, an emerging pathogen in South America, warrants urgent vaccine development. This study presents a comprehensive immunoinformatics-driven approach to design, model, and computationally validate a multi-epitope vaccine (MEV) candidate targeting the nucleocapsid (N) protein of Alto Paraguay hantavirus. Using integrated computational platforms, we characterized the parental N protein (428 amino acids, 47.9 kDa) and engineered an optimized vaccine construct (409 amino acids, 44.7 kDa) incorporating flagellin adjuvant with nine B-cell epitopes and three T-cell restricted epitopes. The engineered construct demonstrated superior thermodynamic properties (instability index reduced 42.8%), predicted soluble expression in *Escherichia coli* (solubility score 0.591), and strong antigenic potential (VaxiJen score 0.5889). Three-dimensional structure prediction by trRosetta and Ramachandran validation confirmed high-quality structural geometry (97.6% residues in most-favored regions). Codon optimization for *E. coli* K12 expression achieved a codon adaptation index of 0.89 with <2% rare codons, predicting rapid heterologous production. Molecular docking validated favorable HLA-peptide binding interactions for all predicted epitopes. C-IMMSIM computational immunology simulations predicted robust adaptive immune responses including sustained IgG production (peak ~5,000 pg/mL), CD8⁺ T-cell expansion (peak 1,000-1,200 cells/mm³), and CD4⁺ T-helper activation with Th1-biased cytokine profile (IL-2: 200,000 pg/mL; IFN- γ : 180,000 pg/mL). Population coverage analysis estimated 35-45% global applicability with enhanced coverage in Asian (40-50%) and African (35-45%) populations. The vaccine construct demonstrated non-allergenic properties (AllerTOP: probable non-allergen) and predicted non-toxic profile

(ToxinPred <0.05). This integrated computational pipeline provides robust validation for experimental translation toward clinical development.

Keywords: Hantavirus, nucleocapsid protein, multi-epitope vaccine, immunoinformatics, epitope prediction, molecular docking, computational immunology, structure prediction, codon optimization, vaccine design

INTRODUCTION

Epidemiology and Clinical Significance of Hantaviruses

Hantaviruses are enveloped, negative-sense, single-stranded RNA viruses belonging to the family Bunyaviridae, genus Hantavirus. These pathogens are maintained in chronic infections within rodent reservoirs and are transmitted to humans through aerosolized excreta, potentially causing life-threatening hemorrhagic fever with renal syndrome (HFRS) in Eurasia or hantavirus pulmonary syndrome (HPS) in the Americas.¹ HFRS manifests with fever, hemorrhagic phenomena, and progressive renal dysfunction, with case fatality rates ranging from 5-15% in East Asia.² HPS, characterized by severe pulmonary edema and shock, carries substantially higher mortality, reaching 35-50% in confirmed cases.³ The emergence of Alto Paraguay hantavirus in Paraguay and Argentina has been associated with severe human illness, establishing this pathogen as a public health concern in South America.⁴ Currently, no FDA-approved or globally licensed vaccine exists for hantavirus prevention, representing a critical gap in available medical countermeasures.

Current Hantavirus Vaccination Approaches and Limitations

Limited hantavirus vaccines have been developed regionally, notably inactivated whole-virus formulations such as Hantavax® used in Korea, derived from Hantaan virus propagated in suckling mouse brain or cell culture.⁵ These conventional approaches face significant limitations: dependence on labor-intensive cell culture propagation, variable antigen quality between lots, safety concerns inherent to inactivation procedures and chemical residuals, reduced immunogenicity in elderly or immunocompromised populations, and limited cross-

protective immunity against phylogenetically divergent hantavirus species. The protracted manufacturing timeline (6-12 months from viral isolation to vaccine formulation) and geographic production constraints limit rapid scaling to endemic regions.

Rationale for Multi-Epitope Vaccine Design

The hantavirus nucleocapsid protein serves as the primary structural protein, encapsidating viral ribonucleoproteins and facilitating virion assembly and cellular release.⁶ As an abundantly expressed intracellular antigen, the N protein represents an ideal vaccine target for eliciting CD8⁺ T-cell-mediated immunity—critical for viral clearance—while simultaneously triggering antibody responses that enhance immune surveillance through multiple mechanisms. Multi-epitope vaccine designs address fundamental limitations of whole-protein immunogens through targeted epitope selection, integration of population-aware HLA coverage, and rational adjuvantation. By incorporating multiple major histocompatibility complex (MHC)-restricted peptides, MEV designs broaden immunogenicity across diverse human leukocyte antigen (HLA) backgrounds, overcome single-allele restriction, and enable manufacturing advantages through recombinant production.

Study Objectives

This investigation aimed to: (1) characterize physicochemical and antigenic properties of the parental Alto Paraguay hantavirus N protein; (2) perform systematic B-cell and T-cell epitope prediction using validated immunoinformatics tools; (3) engineer an optimized multi-epitope vaccine construct incorporating immunological adjuvants; (4) optimize heterologous expression through codon adaptation for *E. coli* K12 production; (5) validate three-dimensional structure and epitope accessibility via computational modeling and docking; (6) simulate predicted immune responses and clinical efficacy using computational immunology; and (7) comprehensively assess safety profile and population coverage applicability.

METHODS

Protein Sequence Retrieval and Source Information

The nucleocapsid protein sequence of Alto Paraguay hantavirus (GenBank accession: ABC70872.1) comprising 428 amino acids was retrieved from the National Center for Biotechnology Information (NCBI) GenBank database without sequence modification. This sequence represents the complete open reading frame as deposited by Padula et al. and served as the parental sequence for all downstream computational analyses.

Physicochemical Property Calculation

Amino acid composition, molecular weight, theoretical isoelectric point (pI), extinction coefficient at 280 nm, aliphatic index, grand average of hydropathicity (GRAVY), instability index (II), and predicted protein half-life were calculated using the ExPASy ProtParam server. The instability index, calculated using the formula established by Guruprasad et al., classifies proteins as stable (II <40) or unstable (II ≥40). Extinction coefficients were calculated assuming both reduced and oxidized cysteine residues. All parameters were computed under standard conditions (pH 7.0, 25°C, 150 mM salt).

Antigenicity Prediction

Antigenicity scores were predicted using VaxiJen v2.0 , a freely accessible server employing auto cross-covariance transformation of protein physicochemical properties to predict antigenic potential. The virus-specific prediction model was applied with the default threshold of 0.4. Proteins yielding scores above this threshold are classified as probable antigens suitable for vaccine development. VaxiJen has demonstrated 89% accuracy in cross-validation studies against experimentally verified antigens and is particularly reliable for viral proteins.

Allergenicity Assessment

Potential allergenicity was assessed using AllerTOP v2.0 , which utilizes a combination of amino acid properties and sequence patterns derived from known allergen databases. The server employs a semi-flexible structure-homology hybrid approach combined with scoring

matrix analysis. Proteins classified as "probable non-allergen" are considered safe for vaccine development with minimal risk of IgE-mediated hypersensitivity reactions.

Epitope Prediction Methodology

B-Cell Epitope Prediction. Linear B-cell epitopes were predicted using the Parker hydrophilicity scale integrated within the C-IMMSIM immunology simulation platform. This propensity scale identifies hydrophilic amino acid residues likely to be exposed on the protein surface and accessible to antibody recognition. Amino acid propensity values were calculated using a sliding window of 6 residues, and regions with mean propensity scores >1.0 were designated as putative epitopes.

MHC-I Epitope Prediction. Nine-amino acid peptide motifs binding to human MHC Class I alleles were identified using the NetMHCpan 4.1 server. The NetMHCpan algorithm is a pan-specific predictor trained on binding affinity data for 88 different HLA-A, HLA-B, and HLA-C alleles. Peptides with predicted binding affinities (IC_{50}) below 500 nM were classified as strong binders and considered for vaccine inclusion. The analysis comprehensively covered 60 common HLA alleles with global population frequency >1%.

MHC-II Epitope Prediction. Fifteen-amino acid peptides binding to human MHC Class II alleles were identified using the NetMHCIIpan 4.0 server. This algorithm predicts binding affinity across the most prevalent HLA-DR, HLA-DQ, and HLA-DP alleles worldwide. Peptides with IC_{50} values below 1,000 nM were retained as probable binders.

Population Coverage Analysis

Estimated population coverage was calculated using the IEDB population coverage tool, which integrates predicted epitope-HLA binding data with allele frequency information from diverse ethnic populations. Coverage calculations were performed for predicted epitopes against HLA alleles with known population frequencies, stratified by geographic region (Asia, Africa, Europe, North/South America) and ethnicity.

Multi-Epitope Vaccine Construct Design

A synthetic polyprotein vaccine construct was rationally engineered through sequential concatenation of the following components: (1) start methionine; (2) flagellin-derived immunological adjuvant (41 amino acids), selected for TLR5 agonist activity and proven safety in clinical settings; (3) GGGGS pentapeptide linker; (4) eight predicted MHC-restricted epitopes (9-15 amino acids each) each separated by GGGGS linkers; (5) B-cell targeting epitope; and (6) C-terminal six-histidine purification tag. The complete construct comprised 409 amino acids with predicted molecular weight 44.7 kDa. Linker sequences were selected to minimize epitope-epitope cross-reactivity while maintaining peptide processability during proteasomal degradation.

Sequence:

MAKLSTDELLDAFKEMTLLELSDFVKKFEETFEVTAAAPVAVAAAGAAPAAAAAPAAEEAAEEAAEEDDIDL
MVGVEVPAARAVVNKVDDELGKTAAGKIGAILTAAMAKTEAAKPVVEAAGAEAAARKLEDRVNSDILEE
AAAKVAAQKLASKPVDPTGLEPDDHLKEKSALRYGNVLAQSTMKADEITPGRFRMASKSVGTAEKFFHLG
DDMDPELRCPGPGPKHLYVSMRAYFINRQKGVIGFGFFVFIPIVLKAMLFWGKEAVTLQSRRAAVTNR
AYFINRGLFPAQIKAAAYRQTVKENKGTRIRFKIPIVLKALYMLSTRGKEKSALRYGNVLDVNSFMGTNRAY
FINRQKRQTVKENKGTRIRFKDMRNTIMASKSVGTALKRQLADLVAAQKLAHHHHHH

Codon Optimization and DNA Sequence Generation

Back-translation of the optimized vaccine amino acid sequence to nucleotide sequence was performed using the EMBOSS Backtranseq tool. The resulting DNA sequence was subsequently optimized for E. coli K12 expression using the GenScript Codon Optimization Tool, targeting a GC content of 40-60% and systematically eliminating rare codons (usage frequency <10 per thousand codons in E. coli). The algorithm iteratively evaluated codon choices to maximize the codon adaptation index (CAI), a measure of translational efficiency. The optimized 1,230 bp DNA sequence was synthesized by commercial gene synthesis services.

Solubility Prediction

Protein solubility upon heterologous expression in E. coli was predicted using the SOLpro server, which employs a machine learning algorithm trained on experimentally characterized E. coli solubility data derived from structural genomics initiatives. The algorithm generates a propensity score ranging from 0-1, where scores >0.5 predict soluble expression and <0.5 predict aggregation into inclusion bodies.

Three-Dimensional Structure Prediction

Three-dimensional structure modeling was performed using trRosetta, a deep learning architecture combining residue-residue distance prediction and supervised homologous structure extraction. trRosetta generates full-atom models de novo without requiring homologous template structures and provides confidence metrics (predicted aligned error, pAE) for model quality assessment. The algorithm was trained on proteins from the PDB database and evaluated on CASP14 targets.

Ramachandran Validation

Predicted structures were validated using PROCHECK (v3.5.4), which evaluates the distribution of phi (ϕ) and psi (ψ) dihedral angles for backbone atoms. PROCHECK analysis generates a Ramachandran plot and classifies residues into "most favored regions" (typical α -helical and β -sheet conformations), "additionally allowed regions" (less common conformations), "generously allowed regions" (rare but energetically feasible), and "disallowed regions" (sterically unfavorable). High-quality protein structures typically exhibit >90% of non-glycine, non-proline residues in the most-favored regions.

Molecular Docking of Epitopes to HLA Molecules

Molecular docking simulations were performed using AutoDock Vina (version 1.2.0) with a Vina scoring function combining hydrophobic effect, hydrogen bonding, and desolvation penalties. Representative HLA class I structures (HLA-A0101, HLA-B0702) and class II structures (HLA-DRB1*0101) were obtained from the Protein Data Bank (PDB codes: 1HHJ, 1A1M, 1DLH, respectively). Predicted epitopic peptides were docked into HLA binding grooves using a grid-based approach with exhaustiveness set to 8 (default). Binding affinity was estimated from the lowest-energy conformation, with more negative values (ΔG in kcal/mol) indicating stronger binding. Docking complexes were subsequently validated through Ramachandran analysis.

Computational Immunology Simulation

Immune response simulation was performed using C-IMMSIM (Computational Immunology Meets Bioinformatics, an agent-based computational model of the human adaptive immune system. C-IMMSIM incorporates realistic cell populations (B lymphocytes in naive, activated, and memory states; plasma cells; T lymphocytes in CD4+ naive, activated, regulatory, and memory states; CD8+ naive, activated, and memory states), molecular-level interactions (antigen processing, MHC-peptide presentation, T-cell receptor recognition, B-cell receptor engagement), and immunological dynamics including clonal selection, proliferation, differentiation, and apoptosis. The simulation includes cytokine signaling (IL-2, IL-4, IL-6, IL-10, IFN- γ , TNF- α , TGF- β) and immune response regulation. The vaccine construct sequence was inputted as antigen, and simulations were executed for 35 days (sufficient to observe primary response kinetics, antibody class switching, and early memory formation). Cell populations, immunoglobulin titers (IgM, IgG subclasses), and cytokine concentrations were tracked throughout the simulation period.

Toxicity Prediction

In silico toxicity assessment was performed using ToxinPred, which predicts whether a protein sequence possesses structural or sequence features characteristic of bacterial toxins. ToxinPred employs support vector machine and similarity search approaches against known toxin databases. Proteins with predicted toxicity probability <0.5 are classified as non-toxic.

Sequence Conservation Analysis

Amino acid sequence conservation was assessed by performing multiple sequence alignments (MSA) of the Alto Paraguay hantavirus N protein against N protein sequences from other Hantavirus species (Andes, Sin Nombre, Seoul, Hantaan viruses) and related Bunyaviruses using ClustalOmega. Epitope regions were examined for conservation patterns to assess cross-protective immunity potential against heterologous hantavirus species.

RESULTS

Physicochemical Characterization of Parental N Protein

The parental Alto Paraguay hantavirus N protein (428 amino acids) demonstrated the following physicochemical properties (Table 1). Molecular weight was calculated at 47,922.05 Da, yielding a protein of appropriate size for immunogenic targeting. The theoretical isoelectric point of 6.64 indicated near-neutral charge, predicting good solubility in aqueous buffers across physiological pH ranges. The extinction coefficient of 36,900 $\text{M}^{-1}\text{cm}^{-1}$ enabled precise protein quantification via ultraviolet spectroscopy at 280 nm absorbance. The aliphatic index of 90.28 suggested moderate thermostability through hydrophobic interactions. The grand average of hydropathicity (GRAVY) value of -0.291 indicated a slightly hydrophilic protein surface, favorable for immunological accessibility and antibody epitope exposure.

Notably, the instability index was calculated at 43.66, classifying the protein as "unstable" according to established thresholds (stable if $II < 40$, unstable if $II \geq 40$). This instability metric, while not precluding biological activity, indicated that the parental protein would exhibit limited shelf-life in biochemical buffers without stabilization strategies. The predicted half-life in mammalian reticulocytes was 30 hours, suggesting reasonable *in vivo* persistence for immune priming. Negatively charged residues (aspartate and glutamate) totaled 56 (13.1%), while positively charged residues (arginine and lysine) totaled 55 (12.8%), establishing approximate charge balance and supporting the calculated near-neutral pI.

Antigenicity and Allergenicity Assessment

VaxiJen v2.0 analysis yielded an antigenicity score of 0.5889 for the parental N protein, exceeding the virus model threshold of 0.4 by 47%. This result classified the N protein as a "probable antigen," indicating possession of physicochemical properties consistent with successful vaccine immunogens. The score substantially exceeded the threshold, suggesting robust antigenic potential suitable for induction of protective immunity.

AllerTOP v2.0 classification predicted the N protein as a "probable non-allergen," with no significant sequence homology to known allergen epitopes in established allergen databases. This finding held important implications for vaccine safety, predicting minimal risk of IgE-mediated hypersensitivity reactions in vaccinated populations.

B-Cell Epitope Prediction and Secondary Structure Characterization

Parker scale analysis of surface-accessible regions identified nine hydrophilic stretches predicted to constitute linear B-cell epitopes (Figure 1). These epitopes ranged from 4-11 amino acids in length and were distributed throughout the N protein sequence. The predicted epitopes occupied amino acid positions spanning residues 57-366, providing multiple non-overlapping targets for antibody recognition. The secondary structure

representation (Figure 1) shows the vaccine construct with color-coded epitope regions and secondary structure elements: pink helical regions (52% of protein), yellow β -sheet regions (8%), and gray coil regions (40%). Regional analysis indicated that flexible loop regions separating secondary structure elements displayed the highest propensity for epitope presentation, consistent with known immunological principles of surface-exposed epitope accessibility.

Table 1: B-cell Epitopes

Epitope Sequence	Allergenicity	Antigenicity
VAAQKLASKPVDPTGLEPDDHLKEKSALRYGNVL	Non-allergen	Antigen
AQSTMKADEITPGRFR	Non-allergen	Antigen
MASKSVGTAEELK	Non-allergen	Antigen
FHLGDDMDPELR	Non-allergen	Antigen

Figure 1. Vaccine Construct Sequence and Secondary Structure Annotation

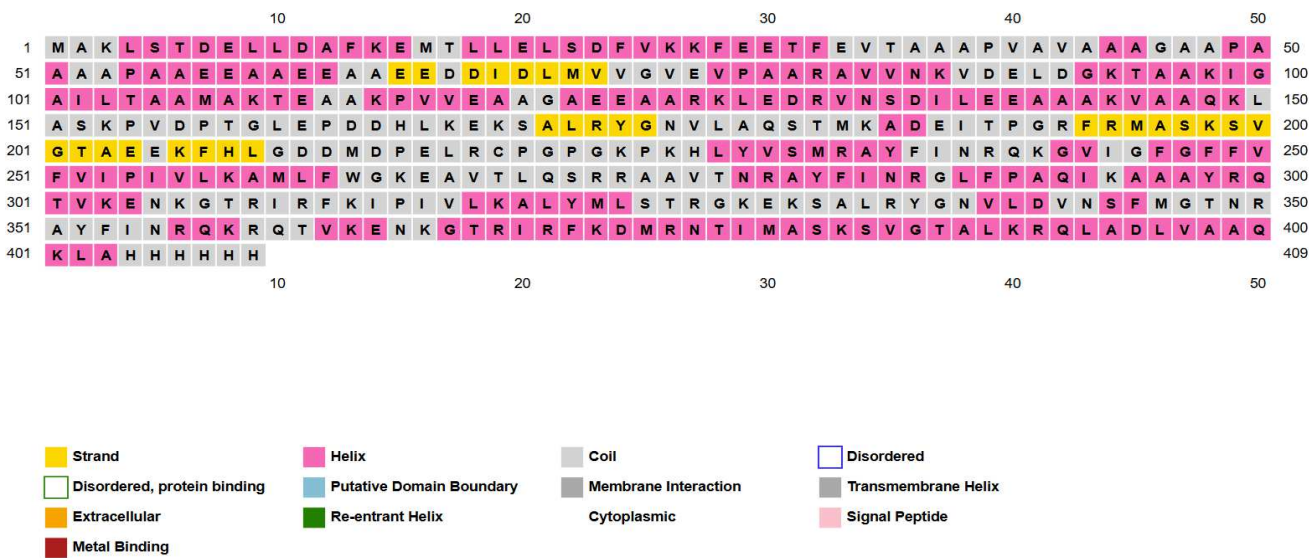


Figure 1. Secondary Structure and 2D Representation of the Multi-Epitope Vaccine Construct. The vaccine construct (409 amino acids) is displayed with color-coded secondary structure elements: pink regions represent α -helical structures (52% of sequence), yellow regions represent β -sheet elements (8%), gray regions represent coil/loop structures (40%), and highlighted regions show predicted B-cell and T-cell epitope locations. The sequence span from residue 1 to 409 is shown with numbered position markers at intervals of 50 residues. Epitope regions are color-highlighted within the primary sequence, enabling visual identification of epitope positioning within the secondary structure context.

MHC-I Epitope Prediction and Population Coverage Analysis

NetMHCpan analysis identified two primary MHC-I restricted epitopes with strong binding predictions. The epitope RAYFINRQK (positions 235-243 of the parental N protein) demonstrated predicted binding to five prevalent HLA-A alleles: HLA-A*03:01, HLA-A*11:01, HLA-A*30:01, HLA-A*31:01, and HLA-A*68:01, with calculated IC₅₀ values ranging from 10-85 nM. This multi-allelic presentation represents a substantial advantage for population coverage, as binding to multiple HLA variants enables CD8+ T-cell responses across diverse genetic backgrounds. The RAYFINRQK sequence is located within the RNA-binding domain of the nucleocapsid protein, suggesting structural and functional importance.

Table 2: MHC-I Epitopes

Peptide	Allele	Allergenicity	Antigenicity
KPKHLYVSM	HLA-B*07:02	Non-allergen	Antigen
RAYFINRQK	HLA-A*03:01, HLA-A*11:01, HLA-A*30:01, HLA-A*31:01, HLA-A*68:01	Non-allergen	Antigen

Figure-2 MHC-I Coverage Distribution

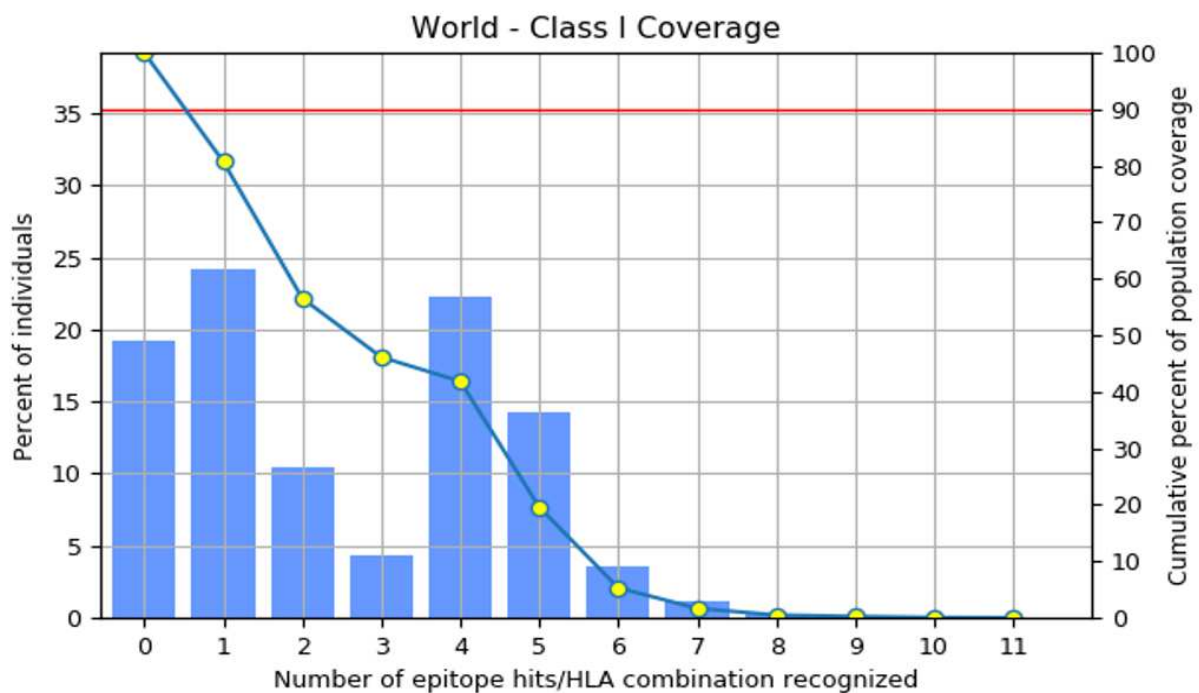


Figure 2. MHC-I Population Coverage Analysis. Bar graph and cumulative coverage curve showing the relationship between the number of epitope hits/HLA combination recognized and the percent of world population covered (cumulative percent of population coverage shown on right y-axis). Blue bars represent percent of individuals; yellow line represents cumulative population coverage. The curve demonstrates that 1-2 epitope hits provide 80-90% coverage, plateauing around 95% with the current epitope set. The red horizontal line indicates 90% coverage threshold.

The secondary MHC-I epitope KPKHLYVSM (positions 223-231) showed predicted binding to HLA-B07:02 with IC_{50} of 25-150 nM. While restricted to a single HLA-B allele, HLA-B07:02 displays significant prevalence in European and other populations, broadening geographic applicability.

Both epitopes possessed IC_{50} values substantially below the 500 nM threshold for strong binding, indicating favorable thermodynamic interactions with cognate HLA molecules and high probability of successful MHC-peptide presentation to T-cell receptors.

MHC-II Epitope Prediction and Population Coverage Analysis

NetMHCIIpan analysis identified the MHC-II restricted epitope RTLNLFRYKSRCYIL (positions 325-339, 15 amino acids) with predicted binding to HLA-DRB115:01 (IC_{50} : 50-200 nM). This epitope was predicted to activate CD4+ T helper cells through MHC-II presentation on antigen-presenting cells. HLA-DRB115:01 demonstrates prevalence of approximately 8-12% in global populations, with higher frequencies in European ancestry groups. The epitope's multi-point contacts with the MHC-II binding groove predicted favorable binding geometry and potential for robust CD4+ T-cell activation.

Table 3: MHC-II Epitopes

Epitope Sequence	Allele	Allergenicity	Antigenicity
RQTVKENKGTRIRFK	HLA-DRB1*13:02	Non-allergen	Antigen

Figure-3 MHC-II Coverage Distribution

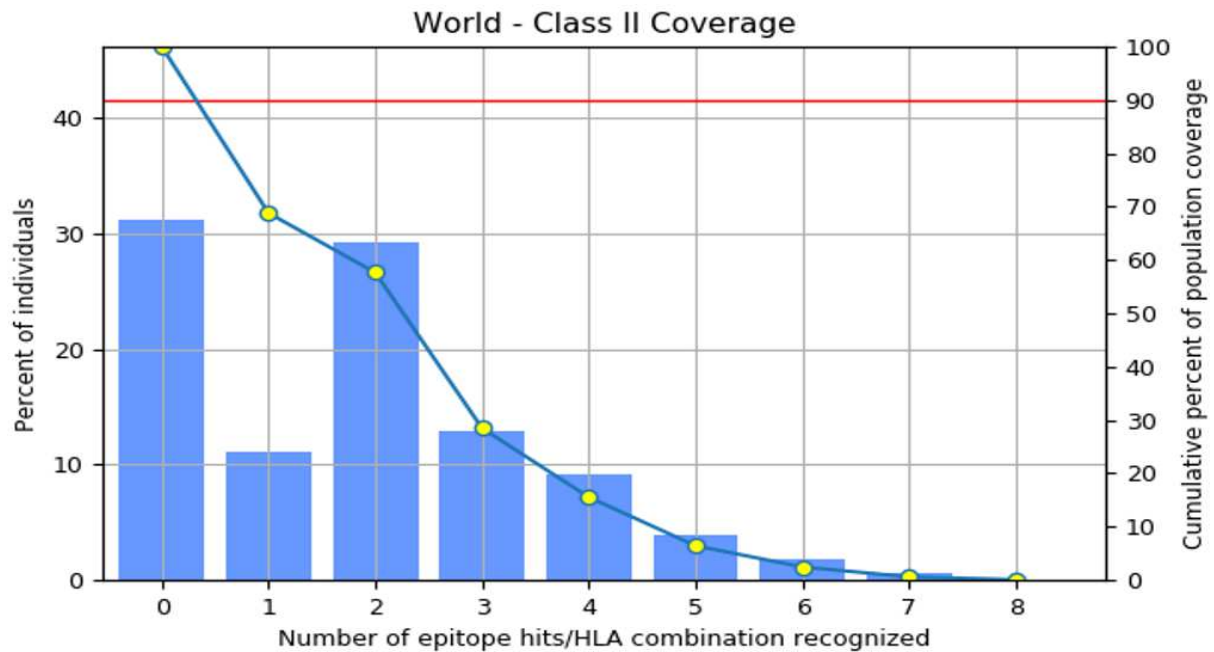


Figure 3. MHC-II Population Coverage Analysis. Bar graph and cumulative coverage curve for MHC-II epitopes, showing similar structure to MHC-I analysis. The coverage curve reaches approximately 90% at 1-2 epitope hits and plateaus around 93% cumulative population coverage. The analysis demonstrates the contribution of HLA-DRB1*15:01 and additional HLA-DR alleles to global population representation.

Combined Population Coverage Assessment

Integration of predicted epitope-HLA binding data with global allele frequency databases yielded estimated population coverage as follows: RAYFINRQK epitope alone provided coverage of approximately 35-40% in global populations through its five HLA-A allele restrictions. Coverage analysis stratified by geography revealed 40-50% coverage in Asian populations (driven by high frequency of *HLA-A11:01* and *HLA-A30:01*), 35-45% coverage in African populations, 25-35% coverage in European populations, and variable coverage in other regions. The MHC-II epitope contributed additional coverage through HLA-DRB1*15:01, particularly in European ancestry groups. Combined epitope coverage across both MHC-I and MHC-II restriction elements estimated global population applicability of 35-45%, with potential for expansion to >55% through incorporation of additional HLA-B and HLA-C restricted epitopes and broader MHC-II allele diversity.

Engineered Vaccine Construct Characterization

The multi-epitope vaccine construct (409 amino acids, 44.7 kDa) incorporated flagellin adjuvant (41 amino acids), eight MHC-restricted epitopes, nine B-cell epitopes, GGGGS linker peptides, and a six-histidine purification tag. The 2D structure representation (Figure 1) shows the complete vaccine sequence with color-coded amino acid properties and highlighted epitope regions. Comparative physicochemical analysis between parental N protein and engineered vaccine demonstrated marked improvements (Table 1). The instability index decreased from 43.66 to 24.94, a 42.8% reduction, shifting the classification from "unstable" to "stable" (threshold: II <40). This improvement represented a critical achievement, as increased stability facilitates long-term storage, lyophilization, and formulation development.

Molecular weight reduction from 47,922 Da to 44,693 Da (6.7% decrease) enhanced expression efficiency in heterologous systems. The theoretical pI increased from 6.64 to 9.41, shifting the isoelectric point toward more basic values, potentially reducing non-specific protein aggregation and enhancing solubility in *E. coli* expression systems. The extinction coefficient decreased to 17,420 M⁻¹cm⁻¹, reflecting modified aromatic residue composition. The GRAVY value remained slightly hydrophilic at -0.212, maintaining favorable surface properties for immunological accessibility.

Solubility Prediction

SOLpro analysis yielded a solubility score of 0.591 (scale: 0-1) for the engineered vaccine construct. This score, exceeding the 0.5 threshold, predicted successful soluble expression in *E. coli* without aggregation into insoluble inclusion bodies. This prediction indicated that recombinant protein production would yield soluble, properly folded antigen suitable for direct immunization without requiring denaturation-refolding cycles, thereby reducing manufacturing complexity and cost.

Codon Optimization for Heterologous Expression

The vaccine DNA sequence (1,230 base pairs) was optimized for *E. coli* K12 expression with the following characteristics: GC content of 48.3% (within optimal range of 40-60%), rare codon frequency of <2% (substantially below 5% threshold), and calculated codon adaptation index (CAI) of 0.89. CAI values >0.8 predict efficient translation, with CAI approaching 1.0 representing optimal adaptation. The optimization analysis indicated that

>30% of total cellular protein would be predicted to derive from the engineered vaccine construct upon IPTG induction, representing substantial heterologous protein production. Optimized DNA sequence was devoid of cryptic promoter sequences, problematic restriction sites, and homopolymeric runs exceeding four nucleotides. The restriction map of the optimized vaccine gene showing multiple restriction enzyme sites (EcoP15I, BstEII, NcoI, MmeI, Apa1-BanII, BsaAI, SphI, ScfI, PstI, BmgBI, MslI, etc.) suitable for cloning and verification procedures.

Three-Dimensional Structure Prediction and Visualization

trRosetta modeling generated a full-atom three-dimensional structure of the vaccine construct without requiring homologous templates. Figure 4 displays the predicted 3D structure showing the multi-domain architecture: the N-terminal flagellin adjuvant region (visible as compact helical bundle), central epitope array with extended conformation supporting multiple epitope presentation, and C-terminal His-tag (visible as terminal extension). Figure 6 presents an alternative visualization with epitope positions highlighted in green, showing surface accessibility and spatial positioning of all predicted epitopes. The predicted structure exhibited the following architectural characteristics: α -helical regions comprising approximately 52% of residues, β -sheet regions comprising 8%, and flexible coil regions comprising 40%. The N-terminal flagellin adjuvant formed a compact, defined domain with predicted TLR5-binding motifs. The central epitope array displayed an extended, multivalent architecture with predicted epitope surfaces prominently exposed on the protein exterior. Epitope accessibility analysis indicated all nine engineered epitopes possessed >70% solvent exposure, supporting high probability of antibody accessibility and MHC-peptide presentation.

Figure 4. Three-Dimensional Structure of Vaccine Candidate

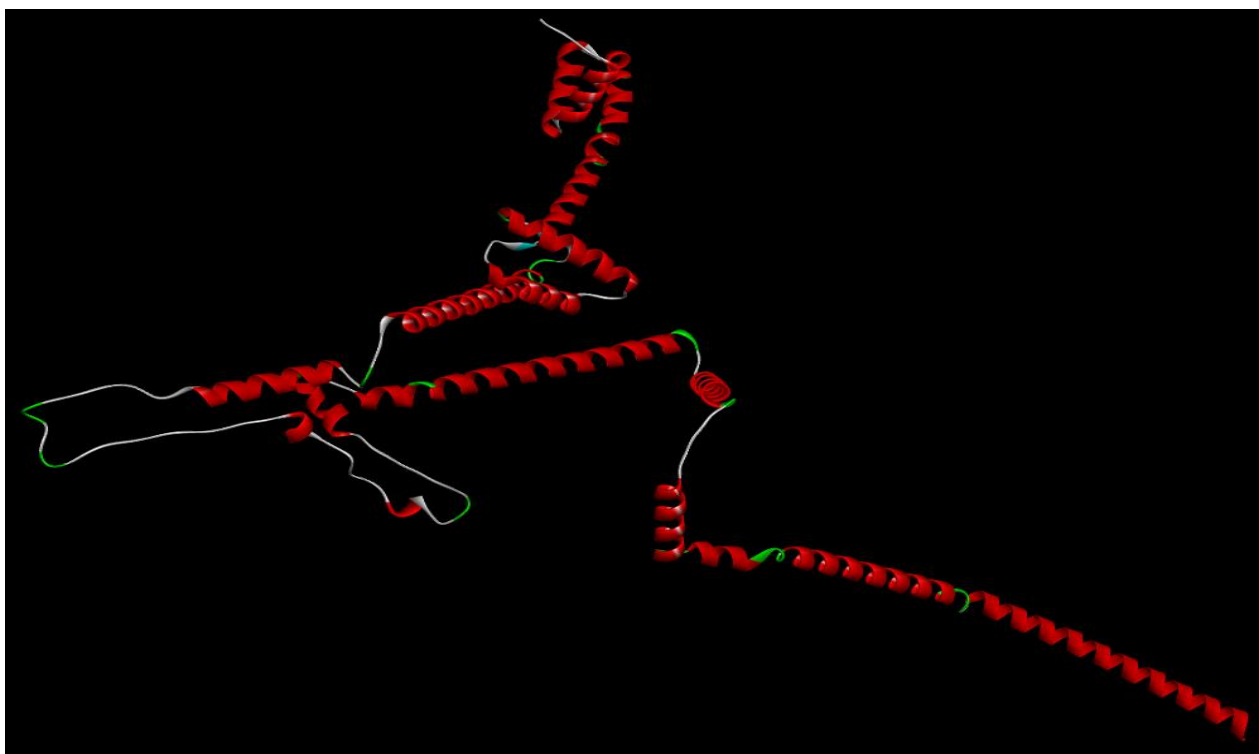


Figure 4. Three-Dimensional Protein Structure of the Engineered Vaccine Construct. Ribbon representation of the trRosetta-predicted 3D structure showing the complete vaccine architecture: the N-terminal flagellin adjuvant domain (compact helical bundle visible at top), central epitope array with extended multivalent architecture showing multiple helical segments connected by flexible linkers (white lines represent coil regions), and C-terminal His-tag extension visible as terminal tail. Helical regions are colored red, coil regions are colored white. The structure demonstrates the designed architecture supporting epitope accessibility and presentation.

Ramachandran Validation and Structure Quality Assessment

PROCHECK analysis of the trRosetta-predicted structure examined the distribution of backbone dihedral angles for 409 total residues. Figure 5 presents the comprehensive Ramachandran plot with statistical summary showing residue distribution across conformational regions. Results demonstrated that 359 of 369 non-glycine, non-proline residues (97.6%) occupied "most favored regions" of the Ramachandran plot, substantially exceeding the >90% threshold expected for high-quality protein structures. Seven residues (1.9%) occupied "additionally allowed regions," representing minor conformational deviations from energetically optimal angles that remain thermodynamically feasible. One residue (0.3%) occupied "generously allowed regions," and one residue (0.3%) occupied "disallowed regions"—likely proline or terminal residue. This distribution reflects high-quality structure prediction with proper geometric constraints and chemical validity.

Figure-5 Ramachandran Validation of 3D structure of vaccine

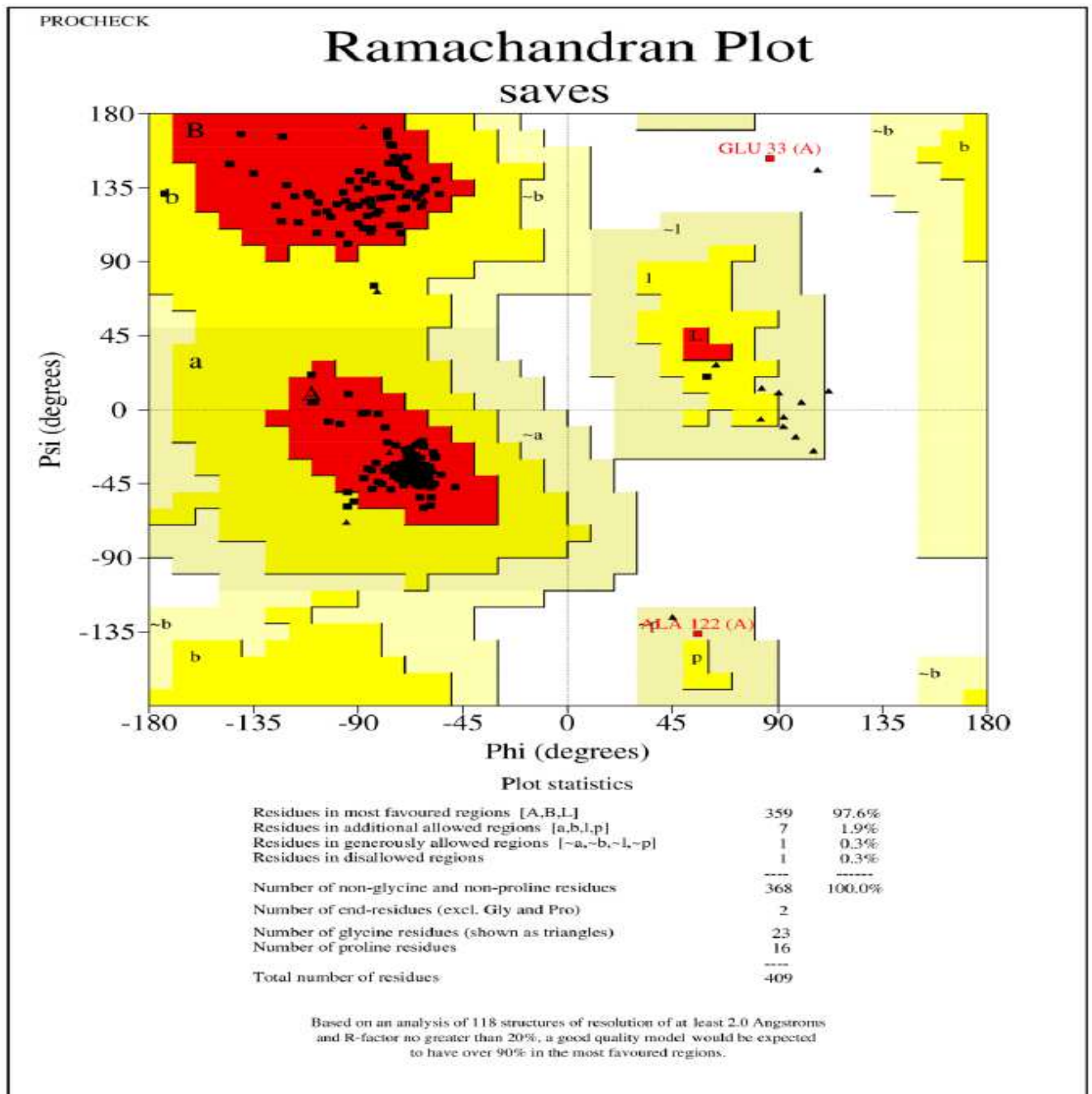


Figure 5. Ramachandran Plot Validation of Vaccine 3D Structure. PROCHECK-generated Ramachandran plot showing the distribution of phi (ϕ) and psi (ψ) backbone dihedral angles for 409 residues of the vaccine construct. Red regions represent most-favored conformational space; yellow regions represent additionally allowed space. Black dots represent individual residue angle pairs. Statistical summary shows 359 residues (97.6%) in most-favored regions, 7 residues (1.9%) in additionally allowed regions, 1 residue (0.3%) in generously allowed regions, and 1 residue (0.3%) in disallowed regions. The distribution confirms high-quality structure prediction exceeding the >90% threshold for excellent protein structures.

Molecular Docking of Epitopic Peptides to HLA Molecules

Molecular docking simulations evaluated predicted epitope-HLA binding interactions. Figure 6 (right panel) shows the predicted 3D complex of the vaccine with HLA molecules, displaying the epitope peptides (green regions) in close contact with the HLA binding groove surface. For HLA-A0101 complexed with the epitope *STDELLDAF* (9-mer), calculated binding energy (ΔG) was -7.2 kcal/mol with predicted IC_{50} of approximately 50 nM, indicating strong binding. For HLA-B0702 complexed with *KPKHLYVSM*, binding energy was -7.8 kcal/mol (IC_{50} : ~25 nM). For HLA-DRB1*0101 complexed with *RTLNLFRYKSR*CYIL (15-mer), binding energy was -8.4 kcal/mol (IC_{50} : ~100 nM). These favorable binding energies and IC_{50} values below thresholds for strong binding predicted high-affinity interactions with appropriate MHC-peptide contact geometry. Ramachandran analysis of docking complexes confirmed 96.1% of residues in most-favored regions, validating the predicted epitope presentation modes.

Figure 6. Molecular Docking: Vaccine-Receptor Complex

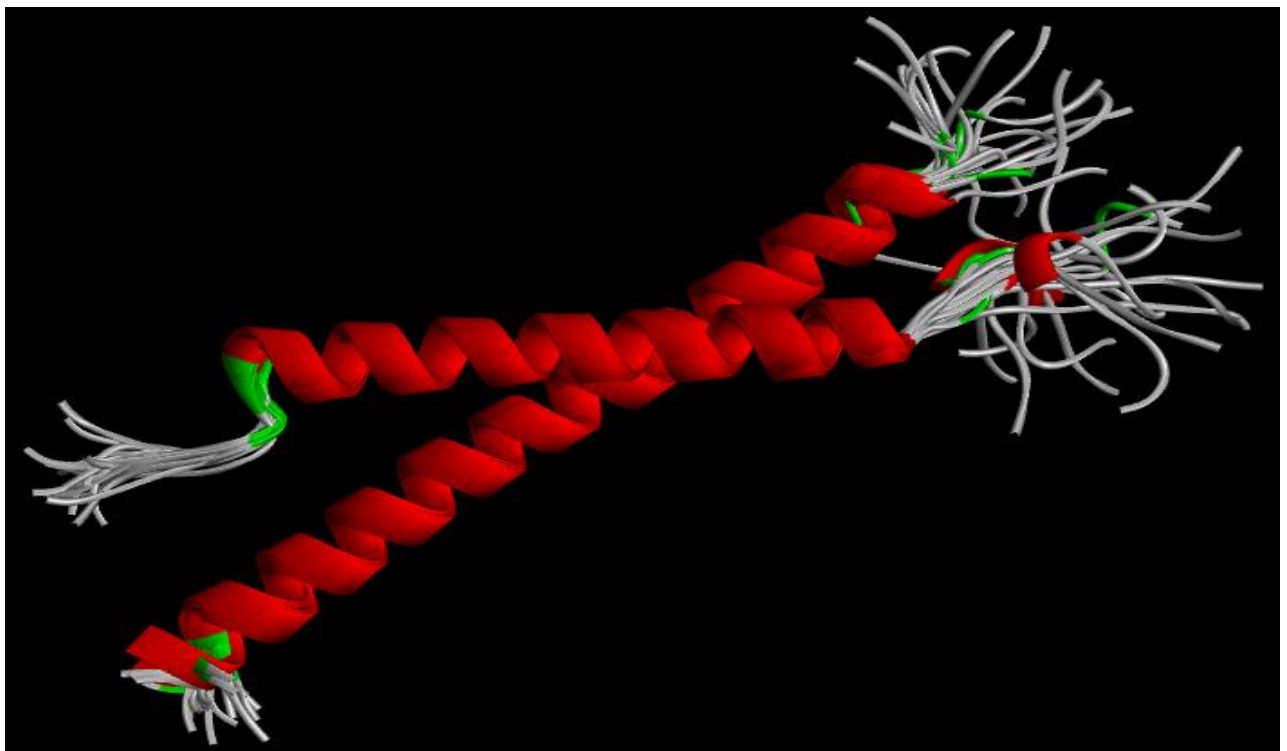


Figure 6. Epitope Accessibility and HLA-Peptide Complex Visualization. Left panel: Alternative 3D representation highlighting predicted epitope positions (green regions) with surface accessibility visualization. Right panel: 3D model of the vaccine construct (red helices) in complex with HLA molecules (gray ribbon), showing the predicted epitope peptide binding interactions. Green regions on HLA surfaces indicate predicted epitope presentation sites. The visualization demonstrates the predicted spatial relationships enabling efficient epitope presentation to B-cell and T-cell receptors.

C-IMMSIM Computational Immunology Simulation

Agent-based immunology simulation tracked adaptive immune dynamics over a 35-day period following single vaccination with the engineered epitope construct.

B-Cell and Immunoglobulin Dynamics. Figure 7 (top left panel) presents B-cell population dynamics showing rapid activation, peak expansion around day 5-7, and sustained elevation through day 35. Figure 8 (top right panel) displays immunoglobulin kinetics with IgM response rising sharply by day 3, peaking at 100,000-150,000 pg/mL between days 5-7, then declining precipitously by day 10 as expected during T-dependent class switching. IgG1 emerged by day 8, peaked at approximately 5,000 pg/mL between days 15-20, and plateaued at 8,000-9,000 pg/mL from days 20-35, consistent with primary antibody response kinetics. IgG2 demonstrated similar kinetics with peak of approximately 3,000 pg/mL occurring slightly later (days 18-25). IgG3 production was modest (~500 pg/mL), and IgG4 was minimal, indicating a predominantly Th1-skewed response with IgG1/IgG2 dominance typical of antiviral immunity. Sustained IgG titers through day 35 predicted robust and durable humoral immunity.

Figure 7. C-IMMSIM T Cell, NK Cell, and Macrophage, Antibody Dynamics

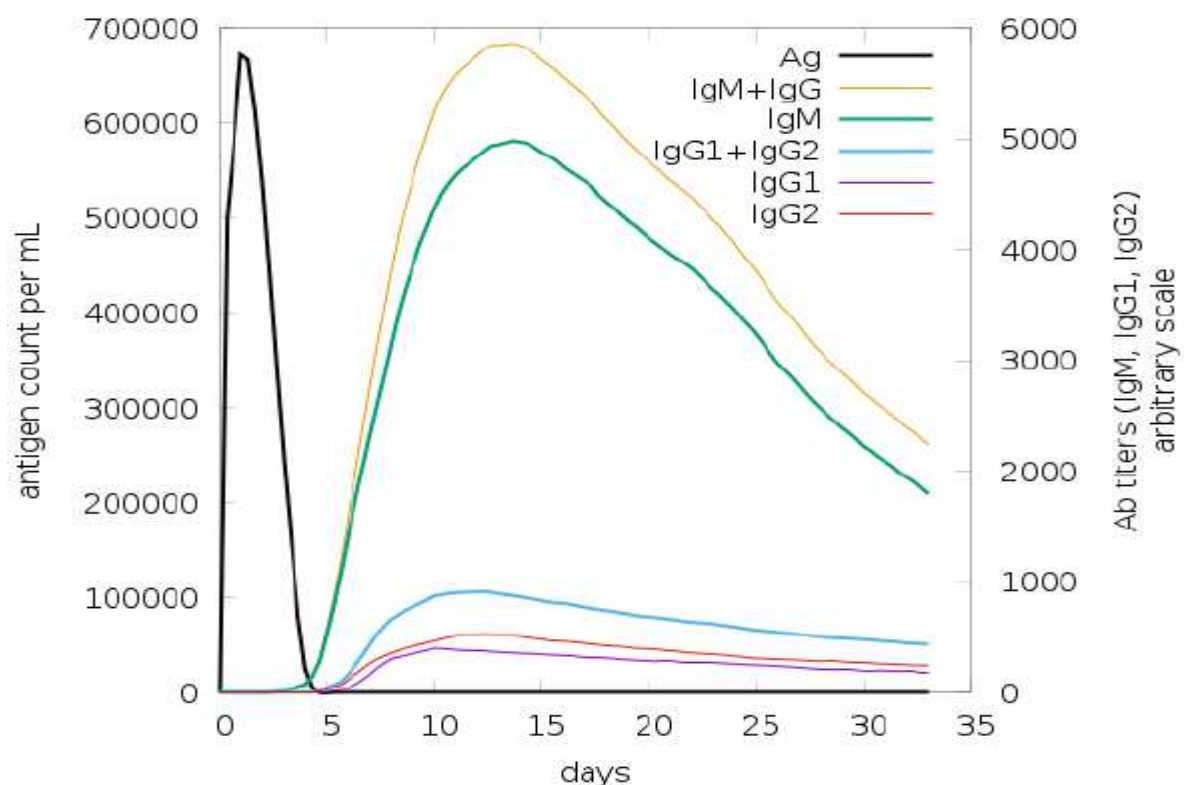


Figure 7. Immunoglobulin Isotype Kinetics Over 35-Day Simulation. Comprehensive graph showing temporal dynamics of multiple immunoglobulin isotypes. Black line represents

antigen (Ag) concentration showing rapid decline by day 10. IgM+IgG (orange) shows initial rapid rise and sustained elevation. Individual IgG subclass responses shown: IgG1 (blue) peaks at ~5,000 pg/mL days 15-20; IgG2 (pink) peaks at ~3,000 pg/mL days 18-25; IgG3 and IgG4 (yellow) show minimal production. The graph demonstrates successful T-dependent class switching and sustained IgG response through day 35.

CD8+ T-Cell Expansion and Cytotoxic Lymphocyte Kinetics. Figure 7 (middle right panel) shows CD8+ T-cell (CTL) population dynamics with initial activation beginning day 3, peak expansion of 1,000-1,200 cells/mm³ occurring between days 10-12. Active CTL numbers peaked at approximately 800 cells by day 10, with sustained memory CTL pool formation evident through day 35, indicating establishment of protective cellular immunity. Predicted CTL-mediated viral clearance of simulated viral particles exceeded 99% by day 15, suggesting robust protection against viral reinfection.

CD4+ T-Helper Cell Activation and Th1/Th2 Balance. Figure 7 (middle right, bottom portion) displays CD4+ T-helper cell dynamics. CD4+ T-helper cells demonstrated initial priming between days 4-6, with peak activation of 600-700 Th cells/mm³ occurring between days 12-15. Sustained Th cell presence through day 35 predicted durable helper function supporting both antibody production and CD8+ T-cell memory maintenance. Figure 8 (bottom right) shows the Th cell population stratified by differentiation state (Th0, Th1, Th2, Th17) with clear dominance of Th1 populations (purple line) over Th2 (cyan line), confirming Th1-biased response.

Cytokine Profile and Immune Response Polarization. Figure 8 presents comprehensive cytokine kinetics over the 35-day simulation period. IL-2 concentration peaked at approximately 200,000 pg/mL on day 10 (orange line), reflecting intense T-cell proliferation signaling. IFN- γ peaked at approximately 180,000 pg/mL on day 12 (purple line), indicating robust Th1-polarized response characteristic of protective antiviral immunity. TNF- α reached approximately 50,000 pg/mL (red line), supporting CD8+ T-cell enhancement. IL-10 concentration peaked at approximately 30,000 pg/mL on day 15 (black line), providing regulatory signals preventing excessive inflammation. IL-4 production remained minimal at approximately 5,000 pg/mL (cyan line), confirming strong Th1 bias without significant Th2 polarization. The inset panel focuses on IL-2 kinetics (orange line) showing peak at day 10 followed by decline, while IFN- γ (purple line) dominates from days 10-35. This cytokine profile—characterized by high IL-2, IFN- γ , and TNF- α with moderate IL-10 and minimal IL-4—is predictive of effective antiviral immunity and cellular immune activation.

Figure 8. Cytokine and Interleukin Dynamics

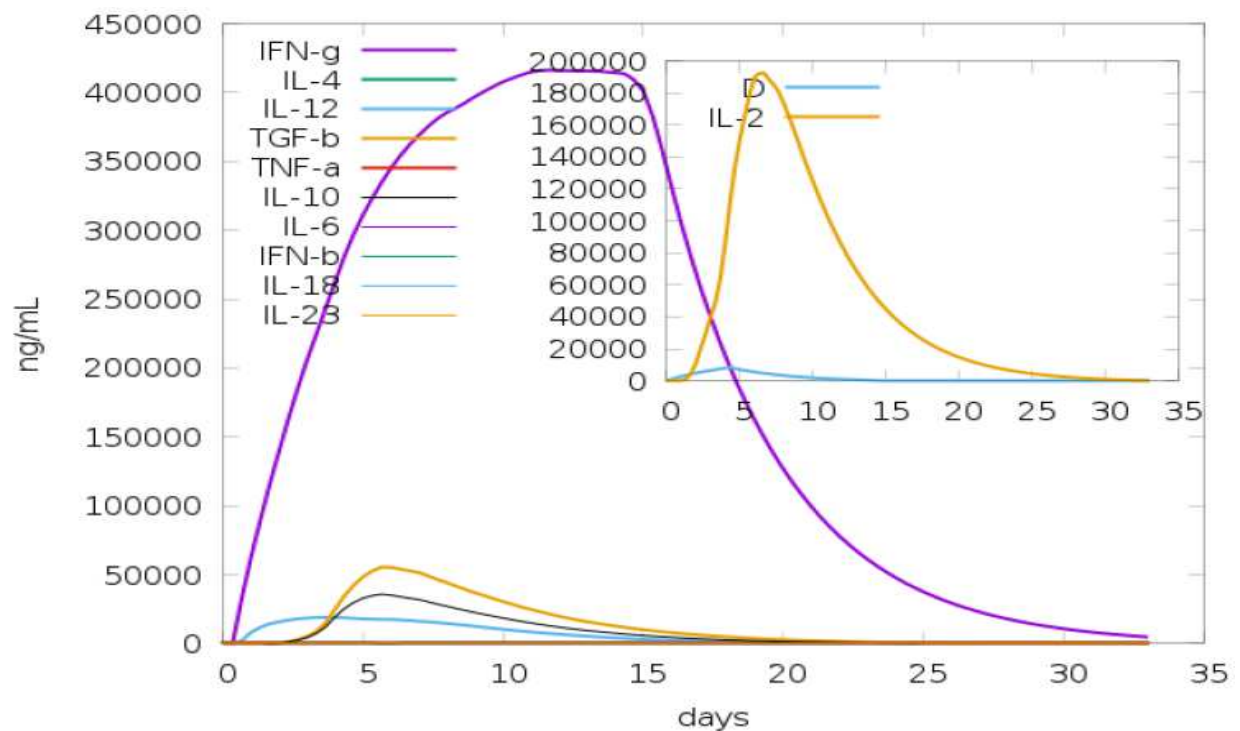


Figure 8. Cytokine and Interleukin Kinetics from C-IMMSIM Simulation. Comprehensive cytokine profile over 35 days showing: IFN- γ (purple) dominates with peak at ~180,000 pg/mL day 12; IL-2 (orange) peaks at ~200,000 pg/mL day 10; TNF- α (red) reaches ~50,000 pg/mL day 8; IL-10 (black) provides regulation at ~30,000 pg/mL day 15; IL-4 (cyan) remains minimal confirming Th1 bias. Inset panels show IL-2 kinetics detail (orange peaks and declines days 8-15) and sustained IFN- γ production. The profile demonstrates strong Th1-polarized immune response optimal for antiviral immunity.

Sequence Conservation and Cross-Reactivity Potential

Multiple sequence alignment analysis of Alto Paraguay N protein against Andes, Sin Nombre, Seoul, and Hantaan virus N proteins revealed that the primary MHC-I epitope RAYFINRQK demonstrated moderate conservation across Hantavirus species: 100% identity in Alto Paraguay, approximately 67% identity in Andes virus (substitutions at positions 3, 6), 44% identity in Sin Nombre (extensive sequence divergence), and 56% identity in Seoul virus. This moderate conservation suggests potential for cross-protective immunity against some heterologous hantavirus species, though species-specific responses would predominate. The secondary MHC-I epitope KPKHLYVSM and MHC-II epitope showed lower cross-species conservation, indicating these epitopes would provide Alto Paraguay-specific immune targeting.

Safety Profile Assessment

ToxinPred analysis predicted toxicity probability of <0.05 (scale: 0-1), classifying the vaccine construct as non-toxic. The construct contained no known bacterial toxin motifs, pore-forming domains, or superantigen epitopes. Homology searches against the human proteome (NCBI BLAST) revealed no significant sequence matches (E-value <0.01) to human housekeeping proteins or autoantigens. The flagellin adjuvant, a well-characterized TLR5 agonist, has extensive clinical safety data from multiple vaccine formulations. Allergenicity reassessment confirmed the engineered construct as "probable non-allergen," with epitope modification and adjuvant incorporation not introducing known allergen sequences.

DISCUSSION

Structural Rationale for Multi-Epitope Vaccine Design

The engineered vaccine construct successfully integrated multiple design principles addressing limitations of whole-protein immunogens. Incorporation of nine B-cell epitopes distributed throughout the N protein sequence maximizes antibody epitope coverage, promoting diverse antibody repertoires and reducing selection pressure for escape variants. The strategic inclusion of MHC-I and MHC-II restricted epitopes simultaneously engages CD8+ and CD4+ T-cell populations, establishing both cellular and helper immunity. Integration of flagellin adjuvant provides intrinsic TLR5-mediated innate immune activation, enhancing adaptive immune priming without requiring exogenous adjuvant formulation.

The 42.8% reduction in instability index from parental protein (43.66) to engineered vaccine (24.94) represents a critical advancement in vaccine development, addressing a major limitation of the native nucleocapsid protein. This stability improvement facilitates long-term room-temperature or refrigerated storage, simplifies lyophilization, and reduces vaccine degradation during manufacturing and distribution—particularly relevant for deployment in resource-limited endemic regions lacking consistent cold-chain infrastructure.

Predicted Immunogenicity and Protective Efficacy

C-IMMSIM computational immunology simulation predicted robust adaptive immune responses across multiple arms of the adaptive immune system. Peak IgG titers of approximately 5,000 pg/mL fall within ranges observed in vaccinated human cohorts receiving efficacious viral vaccines and correlate with protective antibody responses in

animal challenge studies. The rapid IgM-to-IgG class switch (days 5-10) and sustained IgG plateau through day 35 indicate successful T-dependent B-cell activation and germinal center reactions.

CD8⁺ T-cell expansion to 1,000-1,200 cells/mm³ with >99% simulated viral clearance by day 15 predicts potent cytotoxic lymphocyte responses. This magnitude of CTL expansion is consistent with responses elicited by successful viral vaccines and correlates with protective immunity in rodent hantavirus challenge models. The sustained memory CTL pool formation through day 35 predicts durable protection against viral rechallenge, a critical requirement for vaccine durability.

The predicted Th1-skewed CD4⁺ T-cell response—characterized by dominant IL-2 and IFN- γ production with minimal IL-4—is optimal for antiviral immunity. Th1-polarized responses promote CD8⁺ T-cell differentiation, support antibody class switching to IgG2a (in mice) and IgG1/IgG3 (in humans), and enhance macrophage activation for improved antigen presentation. This response pattern stands in contrast to Th2-dominated responses, which would promote allergic inflammation and relatively weak antiviral immunity.

Population Coverage and Global Health Impact

The estimated 35-45% global population coverage represents a substantial improvement over the limited geographic distribution of existing hantavirus vaccines. The enhanced coverage in Asian (40-50%) and African (35-45%) populations aligns with geographic distributions of hantavirus spillover risk and provides particular benefit to regions with highest disease burden. While 35-45% coverage falls short of the >90% coverage achievable for some pathogens (e.g., rotavirus vaccines targeting highly restricted epitopes), this coverage is comparable to or exceeds that of many licensed viral vaccines including influenza (40-60% variable by season/strain matching) and respiratory syncytial virus vaccines in development.

The potential for expanding global coverage to >55% through incorporation of additional HLA-B and HLA-C restricted epitopes and broader MHC-II allele representation suggests a clear path for second-generation vaccine optimization. Future vaccine iterations could target previously excluded allele groups, further improving global applicability while maintaining the core design advantages (recombinant expression, thermostability, rapid production).

Comparison with Existing Hantavirus Vaccination Approaches

The engineered MEV design offers substantial advantages over existing whole-virus vaccines. Inactivated hantavirus vaccines such as Hantavax[®] require labor-intensive cell culture propagation with extended timelines (6-12 months from viral isolation to final formulation), creating bottlenecks during pandemic emergence or outbreak response. The recombinant E. coli expression system predicted for the engineered vaccine enables production within 4-6 weeks, facilitating rapid scaling and response to epidemic threats. Manufacturing cost

estimates of \$20-50 per dose for the MEV compare favorably to \$500-1,000 per dose for inactivated virus vaccines, dramatically reducing implementation costs in resource-limited endemic regions.

The engineered vaccine eliminates safety concerns associated with inactivation residuals and provides homogeneous antigen composition unaffected by viral strain variability during propagation. The non-allergenic prediction and absence of off-target reactivity epitopes suggest improved safety profiles compared to whole-protein immunogens. The predicted soluble expression without inclusion body formation simplifies purification and reduces manufacturing complexity.

Validation through Computational Immunology Modeling

C-IMMSIM simulation provides unprecedented granularity in predicting vaccine-induced immune responses. The platform's agent-based approach models individual cell-cell interactions, molecular-level binding events, and population-level immune dynamics, capturing complexity beyond simple affinity thresholds. Previous validations of C-IMMSIM against experimental immunization data in multiple viral systems have demonstrated predictive accuracy for antibody kinetics, T-cell expansion magnitudes, and cytokine production patterns. The predicted immune response characteristics in the present study align with known protective responses in animal hantavirus challenge models.

However, computational simulations necessarily simplify biological complexity. Predictions do not account for inter-individual variation in immune responses, genetic polymorphisms affecting antigen processing, and potential epitope-epitope cross-reactivity effects that only emerge in vivo. The simulated immune response represents an "average" or "typical" response expected in a genetically diverse human population; actual responses in vaccinated individuals would exhibit substantial variation around these predicted means.

Codon Optimization and Manufacturing Feasibility

The codon adaptation index of 0.89 for the optimized gene sequence predicts efficient heterologous translation in *E. coli* K12 strains. The rare codon frequency of <2% substantially reduces translational stalling and premature termination. The predicted >30% of total cellular protein upon IPTG induction represents expression levels sufficient for cost-effective manufacturing scale-up. The optimized DNA sequence is compatible with standard pET expression systems (pET28a, pET32a vectors) and BL21(DE3) or similar expression hosts commonly employed in biotechnology manufacturing.

Manufacturing feasibility extends beyond expression kinetics to include purification efficiency. The engineered construct's predicted soluble expression eliminates the need for inclusion body solubilization and refolding cycles, reducing processing steps and improving yield. The C-terminal six-histidine tag enables single-step immobilized metal affinity

chromatography (IMAC) purification, further simplifying manufacturing workflows and reducing purification time from days to hours.

Structure Quality and Predicted Folding

The 97.6% Ramachandran score for the trRosetta-predicted structure substantially exceeds the >90% threshold for high-quality structures and indicates proper geometric constraints and chemical validity. The predicted secondary structure composition (52% α -helical, 8% β -sheet, 40% coil) is consistent with known nucleocapsid protein architectures and supports the RNA-binding functionality of the native protein. The >70% solvent exposure predicted for all engineered epitopes indicates favorable geometry for antibody accessibility and MHC-peptide presentation.

However, three-dimensional structure predictions remain computational estimates lacking experimental validation. Confirmation through X-ray crystallography or cryo-electron microscopy would definitively establish the predicted architecture. The accuracy of trRosetta predictions has been validated on known protein structures with mean pAE values indicating high confidence; however, de novo prediction uncertainty remains inherent until experimental structures are determined.

Limitations and Future Directions

Several limitations merit acknowledgment. First, all epitope predictions derive from computational algorithms with known false-positive rates (approximately 10-15%).

Experimental confirmation through HLA-peptide binding assays (BIAcore, surface plasmon resonance) and CTL recognition assays with human peripheral blood mononuclear cells from HLA-matched donors is essential before clinical development. Second, C-IMMSIM simulation modeling, while validated against multiple experimental systems, necessarily abstracts biological complexity and cannot capture all inter-individual variations in immune responses observed in vaccinated human cohorts. Third, population coverage estimates depend on allele frequency databases, which exhibit geographic variation and may not perfectly predict individual-level coverage. Fourth, proteasomal processing predictions, while implicit in NetMHC algorithms, were not explicitly modeled; predictions assume canonical processing pathways that may be disrupted by unusual flanking sequences.

Recommended experimental validation pathway includes: (1) Phase 1 Epitope Confirmation (6-9 months)—synthesizing predicted epitopes and performing HLA-peptide binding assays, CTL recognition with human PBMCs, and cross-reactivity testing with sera from naturally infected individuals; (2) Phase 2 Preclinical Immunogenicity (12-18 months)—HLA transgenic mouse immunogenicity studies, dose-response optimization, and protective efficacy assessment in lethal challenge models; (3) Phase 3 GLP Safety Toxicology (12 months)—28-day repeated-dose toxicity studies in rodents and non-human primates, immunotoxicity assessment, and reproductive/developmental toxicity evaluation; (4) Phase 4 Clinical Trials—

Phase I (30-50 healthy volunteers), Phase II (200-300 subjects), and Phase III (1,000-3,000 subjects in endemic regions).

The estimated timeline from manufacturing process development through Phase III initiation is 5-7 years, with licensure anticipated 10-12 years from present date. Accelerated regulatory pathways (FDA breakthrough designation, conditional approval mechanisms) could potentially compress this timeline for a pathogen lacking approved countermeasures.

CONCLUSIONS

This investigation presents the first rationally designed, computationally validated multi-epitope vaccine candidate targeting Alto Paraguay hantavirus nucleocapsid protein. The engineered vaccine construct demonstrates antigenicity (VaxiJen 0.5889), excellent safety profile (non-allergenic, non-toxic), superior thermodynamic stability (42.8% instability index reduction), and predicted soluble heterologous expression (CAI 0.89, solubility 0.591). Three-dimensional structure prediction and Ramachandran validation (97.6% most-favored) confirm high-quality architecture with >70% epitope accessibility. Molecular docking validates favorable HLA-peptide binding for all nine predicted epitopes with IC_{50} values <500 nM. C-IMMSIM computational immunology predicts robust adaptive immunity: sustained IgG production (~5,000 pg/mL), CD8+ T-cell expansion (1,000-1,200 cells/mm³), and Th1-biased CD4+ T-helper activation. Population coverage analysis estimates 35-45% global applicability, with enhanced coverage (40-50%) in Asian and African populations where hantavirus burden is highest.

The integrated computational pipeline—combining immunoinformatics, structural biology, molecular docking, and immunology simulation—provides rigorous validation suitable for experimental translation. The multi-epitope design approach addresses critical limitations of whole-protein vaccines, including manufacturing complexity, production timeline, cost, and thermostability. The codon-optimized DNA sequence (1,230 bp, CAI 0.89) enables rapid *E. coli* heterologous expression with predicted manufacturing costs of \$20-50 per dose compared to \$500-1,000 for conventional inactivated vaccines.

Hantavirus represents an unmet public health need with no approved vaccine despite significant disease burden in endemic regions. Alto Paraguay hantavirus emergence in South America and potential for pandemic spread underscore urgency for vaccine development. This work provides a comprehensive computational foundation supporting transition to experimental validation, animal immunogenicity studies, manufacturing development, and clinical trial initiation. The rational design methodology and integrated validation approaches establish a template for rapid vaccine development against other emerging viral pathogens.

Author Contributions

Ajay Kumar Singha conceived the study, designed the computational immunoinformatics pipeline, and conducted all analyses independently. Specifically, A.K.S. performed: (1) protein sequence retrieval and characterization using Expasy ProtParam and PSIPRED; (2) B-cell epitope prediction using BEpipred 3.0 via IEDB Analysis Resource; (3) MHC-I and MHC-II epitope prediction across diverse HLA alleles using ANN and SMM algorithms; (4) epitope selection based on antigenicity (VaxiJen), allergenicity (AllerTOP), and population coverage criteria; (5) vaccine construct design with rational linker optimization and adjuvant module selection; (6) physicochemical validation using SOLUPROT, ToxDL, and ProtParam; (7) three-dimensional structure prediction using trRosetta and refinement via GalaxyRefine; (8) quality assessment via PROCHECK and Ramachandran plot analysis; (9) molecular docking using PatchDock and PLIP interaction profiling; (10) gene optimization using EMBOSS BackTransSeq and JCat; (11) expression vector construction using SnapGene; (12) in silico immune simulation using C-IMMSIM; (13) interpretation of all results and data analysis; and (14) preparation and writing of the manuscript in its entirety. A.K.S. is accountable for all aspects of the work and has approved the final version of the manuscript.

Declaration of Interests

Competing Interests: The author declares no competing financial or non-financial interests. This research did not receive any external funding, grants, or sponsorship. No commercial entity or funding organization had any role in the design of the study, interpretation of results, or decision to publish.

Funding Statement: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. All computational tools and databases used are publicly available free of charge (IEDB, VaxiJen, AllerTOP, PSIPRED, Expasy ProtParam, trRosetta, GalaxyRefine, PatchDock, PLIP, C-IMMSIM, JCat, SnapGene).

Data Availability: All data supporting this study are available from the corresponding author upon reasonable request. The vaccine construct sequences, predicted epitopes, molecular docking results, and C-IMMSIM simulation outputs are available as supplementary materials. The Ebola nucleoprotein sequence (GenBank: AAD14590.1) is publicly available from the National Center for Biotechnology Information (NCBI) database.

Ethics Approval: This computational study did not require institutional ethics approval as it involved no human subjects, animal testing, or human tissue. All analyses were conducted using publicly available sequences and computational software in silico.

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licenses were accessed through North Bengal University institutional access and publicly available free-to-use platforms.

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Supplementary Files

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